Lie theory for posets and cones: a gentle introduction

October 2026

Abstract

The free Lie algebra hides inside the partition lattice: the top homology of the lattice of set partitions of $[n]$ is, up to a sign twist, the multilinear part of the free Lie algebra on n letters. This note explains, with small pictures and examples, how that classical fact extends first to finite posets and then to cones of real hyperplane arrangements and of oriented matroids. It describes the fixed-point formula for the characters that appear, says what is known about when those characters vanish, compares the picture with six neighbouring theories, and says what the generalization does and does not buy, including what it says about promotion and cyclic sieving on linear extensions, and one refined count we did not find in the literature (the antichain case is classical): a cyclic order polynomial whose h-vector counts rotation classes of linear extensions by cyclic descents. The poset theorem is now fully formalized in Lean. It is a summary. The proofs are in the notes files named in the footnotes.

How to read this note

Every stated result has a box in the margin and a footnote naming the notes file (and section) it comes from. The boxes mean:

- PROVEN / AUDITED: the source note proves it, and an independent audit re-derived the proof (usually with its own code) and found that it holds. These are the only results stated as *Theorems*. Some of them rest on a cited published theorem; the footnote says which.
- PROVEN / NOT AUDITED: the source note gives a proof, but no independent audit has checked it yet.
- CLASSICAL (CITED): a published result, stated as background.
- KNOWN (INSTANCE): true and audited here, but it is a special case of a published theorem; the footnote names it.
- REFUTED / AUDITED: a conjecture that a source note shows to be false by a counterexample, which an independent audit rebuilt and confirmed.
- COMPUTED: checked by computer on the cases listed, not proved.
- CONJECTURED and OPEN: what they say.

When we say a result was "not found in the literature", we mean that the searches recorded in the notes did not find it. That is not a proof of novelty.

1 The classical story: Lie brackets in the partition lattice

1.1 Lie_n

Take the free Lie algebra on letters $x_1, \dots, x_n$ and keep the part that uses each letter exactly once. Call it Lie_n. It is spanned by bracketings such as $[[x_2, x_3], x_1]$. The symmetric group S_n acts by permuting the letters, and $\dim \mathrm{Lie}_n = (n-1)!$: a basis is the left-normed brackets $[\cdots[[x_1, x_{\sigma(2)}], x_{\sigma(3)}], \dots, x_{\sigma(n)}]$.

There is a second, very concrete model. Let $\gamma = (1\,2\,\cdots\,n)$, let ζ be a primitive nth root of unity, and put

$$e_\zeta = \frac{1}{n} \sum_{j=0}^{n-1} \zeta^{-j} \gamma^j \in \mathbb{C}[S_n].$$

Think of a permutation as a word $w = w_1 w_2 \cdots w_n$. Right multiplication by γ rotates the word, and $w\gamma e_\zeta = \zeta\, w e_\zeta$. So we_ζ depends, up to a root of unity, only on the necklace (rotation class) of w.

Classical fact 1 (Klyachko)**.** $\mathrm{Lie}_n \cong \mathbb{C}[S_n]e_\zeta \cong \mathrm{Ind}_{C_n}^{S_n} \zeta$ as S_n-modules.[1]

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1.2 The partition lattice

Let Π_n be the set partitions of $[n]$, ordered by refinement. The proper part $\overline{\Pi_n}$ removes the bottom (all singletons) and the top (one block). Figure 1 shows Π_3: its proper part is three points, so its reduced homology is $\widetilde{H}_0$, of dimension $2 = (3-1)!$.

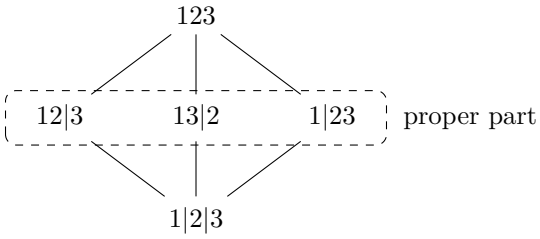


Figure 1: Π_3. The proper part is three points, a wedge of two 0-spheres.

Classical fact 2 (Joyal, Klyachko, Stanley; Hanlon)**.** $\overline{\Pi_n}$ has the homotopy type of a wedge of $(n-1)!$ spheres of dimension $n-3$, and

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$$\widetilde{H}_{n-3}(\overline{\Pi_n}; \mathbb{C}) \cong \mathrm{sgn} \otimes \mathrm{Lie}_n \qquad \text{as } S_n\text{-modules.}$$

Hanlon and Stanley computed the character through fixed points: for g with k cycles all of length d, $\mu(\Pi_n^g) = (-1)^{k-1}\mu(d)\, d^{k-1}(k-1)!$, and $\mu(\Pi_n^g) = 0$ for every other $g \neq 1$.[2]

Here Π_n^g is the sublattice of partitions fixed by g, and μ is its Möbius number $\mu(\hat{0}, \hat{1})$. The reason fixed points appear is the first of the three ideas this note is organized around.

Idea 1 (characters are Möbius numbers of fixed lattices). If the homology of a poset is concentrated in one degree, the Hopf trace formula and P. Hall's theorem turn the trace of a symmetry g on that homology into $\pm\mu$ of the subposet of g-fixed elements. Computing a character becomes computing a Möbius number.

[1] As cited in notes/poset_jks_research_note.tex, §3.1 and notes/poset_jks_theorem.tex, Cor. 3.16(a).

[2] notes/poset_jks_theorem.tex, Cor. 3.16(a); Hanlon's method as read in notes/novelty_search_2.md, §2.1.

2 Posets

2.1 Order-preserving partitions and linear extensions

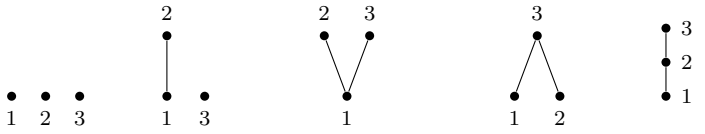
Now replace the n unrelated letters by a finite poset P on $[n]$. A set partition π is *order-preserving* if collapsing each block to a point leaves a directed graph without cycles. These partitions form a lattice $\mathcal{O}(P)$. For the antichain nothing is forbidden and $\mathcal{O}(P) = \Pi_n$. For the chain $1 < 2 < 3$ the partition $13|2$ is forbidden, because $1 < 2 < 3$ would become a cycle between the blocks $\{1, 3\}$ and $\{2\}$.

A *linear extension* of P is a word w in which i comes before j whenever $i <_P j$. Write $\mathrm{Lin}(P)$ for the set of them, and $e_C(P)$ for the number of rotation classes of words that contain at least one linear extension. If P is connected, no two linear extensions are rotations of each other, so $e_C(P) = |\mathrm{Lin}(P)|$.

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Classical fact 3 (Jenča–Sarkoci)**.** For $n \geq 3$, the proper part of $\mathcal{O}(P)$ is homotopy equivalent to a wedge of $e_C(P)$ spheres of dimension $n - 3$.[3]

Figure 2 lists the five posets on three points.



	antichain	$1 < 2,\ 3$	$1 < 2,\ 1 < 3$	$1 < 3,\ 2 < 3$	chain
$\lvert\mathrm{Lin}(P)\rvert$	6	3	2	2	1
$\lvert\mathcal{O}(P)\rvert$	5	5	5	5	4
$e_C(P) = \dim \widetilde{H}_0$	2	2	2	2	1

Figure 2: The posets on three points, by direct enumeration. For the second poset the three linear extensions $123, 312, 132$ form two rotation classes, $\{123, 312\}$ and $\{132\}$. For the chain, $\mathcal{O}(P)$ is the Boolean lattice $\{1|2|3,\ 12|3,\ 1|23,\ 123\}$.

2.2 The poset Lie module

Define

$$\mathsf{Lie}(P) = \mathrm{span}\{we_\zeta : w \in \mathrm{Lin}(P)\} \subseteq \mathbb{C}[S_n].$$

Its dimension is $e_C(P)$, and the automorphism group $\mathrm{Aut}(P)$ acts on it by relabelling letters. For the antichain it is all of $\mathbb{C}[S_n]e_\zeta$, which is Lie_n. Let $H_P = \widetilde{H}_{n-3}(\overline{\mathcal{O}(P)}; \mathbb{C})$.

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Theorem 1 (poset JKS)**.** *Let P be a finite poset on $n \geq 3$ points. Then $H_P \cong \mathrm{sgn} \otimes \mathsf{Lie}(P)$ as $\mathrm{Aut}(P)$-modules. Equivalently, for every $g \in \mathrm{Aut}(P)$,*

$$(-1)^{n-3}\mu(\mathcal{O}(P)^g) = \mathrm{sgn}(g) \sum_{C\,:\,gC=C} \zeta^{r_g(C)},$$

where C runs over the rotation classes meeting $\mathrm{Lin}(P)$ that g maps to themselves, and $r_g(C)$ is the rotation by which g acts on C. Moreover both sides equal $\mathrm{sgn}(g)\,\mu(o) \sum_{x \in F(P,g)} e_C(P|_{B_0(x)})$, where o is the order of g, $\mu(o)$ is the number-theoretic Möbius function, and $F(P, g)$ is the set of

[3] Cited as input (A) in notes/poset_jks_theorem.tex, §2; the statement was checked against the source in notes/citation_check.md, §2.

partitions into o blocks, each a union of connected components, that g permutes cyclically ($B_0(x)$ is any block).[4]

For the antichain this is the classical JKS theorem. The right-hand side of the character identity has a plain meaning: it is a sum over the rotation classes of linear extensions that g fixes, each weighted by a root of unity. At $g = 1$ it is the count $e_C(P)$, and the identity becomes $\mu(\mathcal{O}(P)) = (-1)^{n-1} e_C(P)$. That count also has a proof with no topology at all: one application of Weisner's theorem and Jenča–Sarkoci's recursion for e_C.[5]

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Corollary 2. *If P is connected and $n \geq 3$, then $\mu(\mathcal{O}(P)^g) = 0$ for every $g \neq 1$, and H_P is a free $\mathrm{Aut}(P)$-module of rank $|\mathrm{Lin}(P)|/|\mathrm{Aut}(P)|$.*[6]

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So roots of unity appear only when g permutes connected components freely.

More is true at the level of spaces. For connected P, the order complex of the proper part of $\mathcal{O}(P)$ is a wedge of $e_C(P) = |\mathrm{Lin}(P)|$ spheres that $\mathrm{Aut}(P)$ permutes freely, up to $\mathrm{Aut}(P)$-equivariant homotopy, so H_P is free even over $\mathbb{Z}[\mathrm{Aut}(P)]$.[7]

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Example 1 (a fixed-point computation)**.** Let P be two disjoint chains $1 < 2$ and $3 < 4$, and let $g = (13)(24)$ swap them. The linear extensions are the six shuffles of 12 and 34. The words 1234 and 3412 are rotations of each other, and the other four lie in four further classes, so $e_C(P) = 5$ and $\dim H_P = 5$.

Lie side. Only the class $\{1234, 3412\}$ is fixed by g: g sends 1234 to 3412, which is 1234 rotated by two places. With $\zeta = i$ the phase is $\zeta^2 = -1$. So the trace of g on $\mathsf{Lie}(P)$ is -1.

Möbius side. The g-fixed order-preserving partitions are drawn in Figure 3. Counting upwards, μ is -1 on the three atoms, $+1$ on $13|24$ and $+1$ on the top. So $(-1)^{4-3}\mu(\mathcal{O}(P)^g) = -1$, and $\mathrm{sgn}(g) = +1$. The two sides agree. The closed formula gives the same number: $o = 2$, $\mu(2) = -1$, $F(P, g) = \{12|34\}$, and e_C of a 2-chain is 1.

Why it works (Weisner). The orbit partition of g is $O_g = 13|24$. Weisner's theorem says that the Möbius numbers $\mu(\hat{0}, x)$ of the x with $x \vee O_g = \hat{1}$ sum to zero. Here those x are $12|34$ and $\hat{1}$, so $\mu(\hat{0}, \hat{1}) = -\mu(\hat{0}, 12|34) = 1$. The general proof is this step plus a Möbius inversion.[8]

Remark. The first proposed route to Theorem 1 was to show that one fixed classical JKS isomorphism maps H_P onto $\mathsf{Lie}(P)$. That is false already for a chain and its reverse: they have the same $\mathcal{O}$, but different spans $\mathbb{C}\, we_\zeta$.[9] The correct restricted statement replaces $\mathbb{C}[S_n]e_\zeta$ by $\mathbb{C}[S_n]$ modulo shuffles. That leads to the second idea.

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[4] notes/poset_jks_theorem.tex, §3, Theorem 3.1. Two independent proofs, both audited in notes/audit_character_identity_proof.md ("proof holds"). The proof uses Jenča–Sarkoci, P. Hall's theorem and the Hopf trace formula, all cited; all three, and the theorem itself, are now proved in Lean (§8). The isomorphism is abstract (equal characters); an explicit choice-free isomorphism is in §6 of the same file.

[5] notes/mu_op_elementary.md, Theorem A (Proof W), audited in notes/audit_mu_op_elementary.md: both proofs hold and are homology-free.

[6] notes/poset_jks_theorem.tex, Cor. 3.15; audited in notes/audit_character_identity_proof.md, II7.

[7] notes/theorem_e_equivariant_wedge.md, Theorem E and §2.5; audited in notes/audit_theorem_e.md (it holds). For the claw (one point below all the others), where $\mathcal{O}(P) = \Pi_n$, this is the known splitting of the partition complex under S_{n-1} (Arone; Arone–Brantner; Donau); for other connected P we did not find it (notes/prior_art_theorem_e.md).

[8] notes/poset_jks_theorem.tex, §3.2, Prop. 3.11 and Lemma 3.8 (Weisner). The computation in this example was done by hand and by direct enumeration.

[9] notes/poset_jks_theorem.tex, §4, Prop. 4.2; audited in notes/audit_literature_novelty_jks_maps.md.

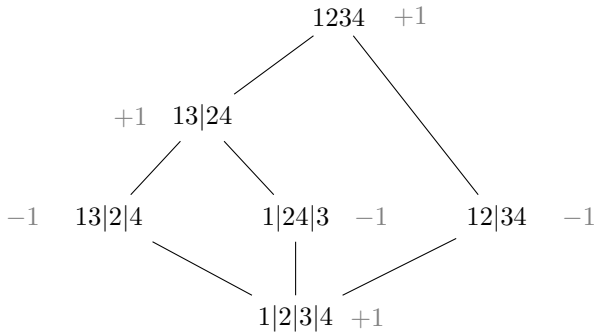


Figure 3: The lattice $\mathcal{O}(P)^g$ for $P = \{1 < 2\} \sqcup \{3 < 4\}$ and $g = (13)(24)$, with $\mu(\hat{0}, x)$ in grey.

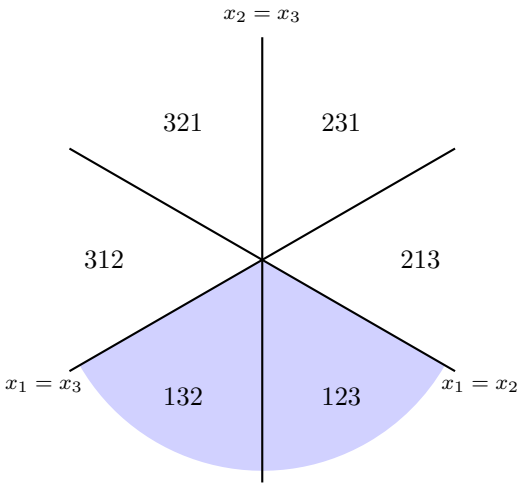
3 Geometry: chambers, cones and lunes

3.1 The braid arrangement

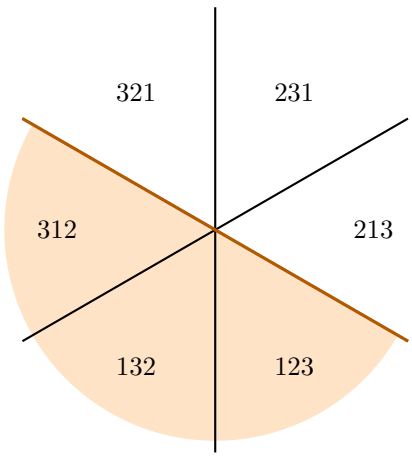
The braid arrangement is the set of hyperplanes $x_i = x_j$ in $\mathbb{R}^n$. Its chambers (the regions where all coordinates differ) are the permutations: the chamber of w is $x_{w_1} < x_{w_2} < \cdots < x_{w_n}$. A poset cuts out the *order cone*

$$C_P = \{x \in \mathbb{R}^n : x_i \leq x_j \text{ whenever } i <_P j\},$$

and the chambers inside C_P are exactly the linear extensions of P. Figure 4 shows $n = 3$, after quotienting by the line $x_1 = x_2 = x_3$: three lines, six chambers.



(a) $P = \{1 < 2,\ 1 < 3\}$: a pointed cone



(b) $P = \{1 < 2\}$: a half-plane, a top-lune

Figure 4: The braid arrangement A_2. (a) The order cone of $1 < 2,\ 1 < 3$ contains the chambers 123 and 132. Its faces span three lines, so its support lattice has three middle elements and $m(V) = 2$. (b) The order cone of $1 < 2$ is a half-plane containing $123, 132, 312$. It contains the line $x_1 = x_2$ (its lineality space), and it is itself a top-lune.

Faces of the arrangement inside C_P correspond to order-preserving maps $P \to [k]$ onto chains, and the linear span of a face is recorded by the partition into fibres. In this way the lattice of spans

of faces of C_P is $\mathcal{O}(P)$, turned upside down.[10]

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3.2 Any arrangement, any cone

Let $\mathcal{A}$ be a finite central arrangement of hyperplanes in $\mathbb{R}^d$, essential (the hyperplanes meet only in 0). A *top-cone* V is a nonempty set of chambers obtained by choosing a side of some of the hyperplanes. Its *support lattice* $\Lambda(V)$ is the set of subspaces spanned by faces of $\mathcal{A}$ that lie in the closed cone $|V|$, ordered by inclusion. Put $m(V) = (-1)^d \mu(\Lambda(V))$. The *lineality space* $\operatorname{lin} V = |V| \cap -|V|$ is the largest subspace inside the cone. The cone is *pointed* if $\operatorname{lin} V = 0$.

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Theorem 3 (cone wedge theorem)**.** *For every top-cone V of a finite real central essential arrangement of rank d, the proper part of $\Lambda(V)$ is homotopy equivalent to a wedge of $m(V)$ spheres of dimension $d-2$.*[11]

For the braid arrangement and $V = C_P$ this is the Jenča–Sarkoci theorem, with a new proof, and $m(C_P) = e_C(P)$. For $V =$ all chambers it is Folkman's theorem for geometric lattices.

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Theorem 4 (Zaslavsky for cones)**.** *$\#V = \sum_X m(V_X)$, where X runs over the flats contained in $\operatorname{lin} V$ and V_X is the image of V in $\mathbb{R}^d/X$. If V is pointed, $\#V = m(V)$.*[12]

In Figure 4(b), $\#V = 3$, $m(V) = 2$, and the line $x_1 = x_2$ contributes 1. In the braid case the theorem reads $|\operatorname{Lin}(P)| = \sum_\sigma \prod_{B \in \sigma} e_C(P|_B)$, a sum over partitions σ of the components of P.

3.3 Lie elements are what lunes cannot see

A *top-lune* is the set of chambers lying on prescribed sides of all the hyperplanes through some fixed flat. In the braid arrangement the top-lunes are the sets of shuffles of a few words, for example $\{123, 132, 312\}$, the shuffles of 12 and 3. Let Γ be the space spanned by the chambers, and let $\operatorname{Lune} \subseteq \Gamma$ be spanned by the sums over top-lunes with more than one chamber. The Lie elements of the arrangement, in the sense of Aguiar–Mahajan, are the vectors orthogonal to Lune. For the braid arrangement Lune is the span of the proper shuffles, and its orthogonal is Lie_n (Ree).[13]

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Idea 2 (chambers modulo lunes). The Lie module of a cone is the span of its chambers, modulo the lunes. The top homology of the cone's support lattice is that module, twisted by the determinant.

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Theorem 5 (cone JKS)**.** *Let V be a top-cone of a finite real central essential arrangement in $\mathbb{R}^d$, $d \geq 2$, and let G be a finite group of linear maps preserving $\mathcal{A}$. Write Γ_V for the span of the chambers of V. Then*

$$\widetilde{H}_{d-2}(\overline{\Lambda(V)}; \mathbb{C}) \cong \det \otimes (\Gamma_V + \operatorname{Lune})/\operatorname{Lune}$$

as modules over the stabilizer of V in G. In particular $m(V)$ equals $\dim \Gamma_V/(\Gamma_V \cap \operatorname{Lune})$.[14]

[10] notes/cone_lune_lie_theory.md, §2, Prop. 2.1 (audited in notes/audit_cone_lune_lie_theory.md), and notes/general_theory.md, §1.3, Cor. 1.4 (audited in notes/audit_general_theory_cones.md).

[11] notes/general_theory.md, §1.1, Theorem 1; audited in notes/audit_general_theory_cones.md (holds; the cited fibre theorem of Björner–Wachs–Welker can be removed, Prop. A there). We did not find it in the literature (notes/novelty_general_theory.md, §2).

[12] notes/general_theory.md, §1.2, Theorem 2; audited in notes/audit_general_theory_cones.md. It is a short specialization of Zaslavsky's 1977 dissection theorem (notes/novelty_general_theory.md, §3).

[13] Aguiar–Mahajan, *Topics in Hyperplane Arrangements*, Lemmas 10.7–10.8, as read in notes/literature_novelty_jks_maps.md, §4.2, and notes/cone_lune_lie_theory.md, §1.

[14] Proved as Theorem C in notes/audit_general_theory_cones.md, §C, given a theorem of Brown on left regular bands (cited, read in the source); restated as Theorem G* in notes/general_theory.md, §1.5. Independently audited in notes/audit_theorem_c.md (holds with fixes to proofs; no conclusion changes). For $V =$ all chambers it is known (Aguiar–Mahajan, *Topics*, Thm. 14.32); for proper cones we did not find it (notes/novelty_general_theory.md, §5).

For C_P the determinant is sgn, and Theorem 5 says $H_P \otimes \mathrm{sgn} \cong \mathbb{C}[\mathrm{Lin}\, P]/(\text{lunes inside } C_P)$. That is the poset theorem in a form that names an explicit basis: the splitting cycles of the linear extensions.[15] In Figure 4(b), three chambers modulo one lune leave a two-dimensional space, and $m(V) = 2$. A pointed cone contains no lune with more than one chamber, and its chambers are independent modulo lunes.

3.4 Oriented matroids

Nothing above needs coordinates. An oriented matroid keeps only the sign vectors of the faces, and some oriented matroids come from no real arrangement. The role of a top-cone is played by a T-convex set of topes.

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Theorem 6. *For every simple oriented matroid of rank r and every nonempty T-convex set V of topes: the proper part of $\Lambda(V)$ is a wedge of $m(V)$ spheres of dimension $r-2$; the Zaslavsky count holds; and the top homology is* $\det \otimes (\Gamma_V + \mathrm{Lune})/\mathrm{Lune}$ *as a module over the stabilizer of V, given Brown. Realizability is not used.*[16]

4 Characters: the fixed-point formula

Let g be a linear map of finite order with $g\mathcal{A} = \mathcal{A}$ and $gV = V$, for instance an automorphism of P acting on C_P. By Idea 1 its character on the top homology is

$$\chi_V(g) = (-1)^d \det(g)\, \mu(\Lambda(V)^g),$$

where $\Lambda(V)^g$ is the lattice of g-stable spans. In the braid case this is the left side of Theorem 1, multiplied by $\mathrm{sgn}(g)$.

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Theorem 7 (equivariant cone Zaslavsky)**.** *The number of chambers of V fixed by g is*

$$\#V^g = \sum_X \chi_{V_X}(\bar{g}),$$

the sum over the g-stable flats $X \subseteq \mathrm{lin}\, V$, where $\bar{g}$ is the map induced on $\mathbb{R}^d/X$. Inverting this over the lattice of g-stable flats gives a closed formula for $\chi_V(g)$ in terms of chambers fixed by g, which can be counted on the fixed space $\mathrm{Fix}(g)$.[17]

At $g = 1$ this is the cone Zaslavsky theorem. On a pointed cone it says $\chi_V(g) = \#V^g$. If $\mathcal{A}$ is the reflection arrangement of a reflection group W and $1 \neq g \in W$, then g fixes no chamber, so the top homology of a pointed cone is a free module over the stabilizer of V in W.[18] This is the analogue of the connected-poset corollary. For the braid arrangement the formula of Theorem 7 agrees with the closed formula of Theorem 1.

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[15] notes/poset_jks_theorem.tex, §5, Theorem 5.3(d), which is proved there given a cited theorem of Wachs; by Theorem 5 it also holds given Brown (notes/general_theory.md, §1.5).

[16] notes/oriented_matroid_cones.md, Verdict table, Theorems 1-OM, 2-OM and C-OM; audited in notes/audit_oriented_matroid_cones.md: every claim holds, with one wrong attribution fixed. For $r \geq 5$ the homotopy form uses shellability of the tope complex (cited).

[17] notes/fixed_point_formula.md, Verdict table, rows EZ and CF; audited in notes/audit_fixed_point_formula.md (holds, with fixes). For V = all chambers both are known in substance (Moseley; Sundaram–Welker); for proper cones we did not find them (notes/novelty_fixed_point_formula.md, §§2–3).

[18] notes/fixed_point_formula.md, Cor. P; audited in notes/audit_fixed_point_formula.md.

5 When does the character vanish?

Idea 3 (transverse flats). Whether $\chi_V(g)$ can be nonzero is decided by the g-stable flats X that meet $\mathrm{Fix}(g)$ only in 0 and are transverse to it (no hyperplane contains $X + \mathrm{Fix}(g)$), and by the Möbius numbers of their fixed lattices.

Write $Z^*(V, g)$ for the g-stable flats $X \subseteq \operatorname{lin} V$ with $X \cap \mathrm{Fix}(g) = 0$ such that no hyperplane of $\mathcal{A}$ contains $X + \mathrm{Fix}(g)$, and $Z^*(g)$ for the same set with $V =$ all chambers.

Theorem 8 (Formula Z and the some-cone criterion)**.** PROVEN AUDITED

$$\chi_V(g) = \sum_{X \in Z^*(V,g)} \mu_{L(\mathcal{A})^g}(0, X)\, m_X(V), \qquad m_X(V) \geq 1,$$

where $L(\mathcal{A})^g$ is the lattice of g-stable flats and $m_X(V)$ is the multiplicity of a smaller cone inside $\mathrm{Fix}(g)$*. Consequently $\chi_V(g) \neq 0$ for some g-stable top-cone V if and only if some $X \in Z^*(g)$ has $\mu(0, X) \neq 0$, and then a top-lune witnesses it.*[19]

Theorem 9 (type A, all of $\pm S_n$)**.** *On the braid arrangement in $\mathbb{R}^n$ ($n \geq 3$), let $g = \pm w$ with $w \in S_n$. Some g-stable cone has $\chi \neq 0$ if and only if $\langle g \rangle$ acts freely on $\{\pm e_1, \ldots, \pm e_n\}$ and $\mu(\operatorname{ord} g) \neq 0$. For $+w$ this is the classical rule "all cycles of one length o, with $\mu(o) \neq 0$". For $-w$ it says that the cycle lengths of w lie in $\{m, 2m\}$ for one odd squarefree m.*[20] PROVEN AUDITED

The map $-w$ preserves an order cone C_P exactly when w is an anti-automorphism of P (it reverses every relation), so the second half is about anti-automorphisms of posets. Since $\mathcal{O}(P) = \mathcal{O}(P^{\mathrm{op}})$, they act on $\mathcal{O}(P)$ too.[21] Already for $n \leq 7$ there are nine classes $-w$ that are nonzero on some cone but zero on all chambers, for example $-(2, 1)$.[22] The rule for $-w$ is invisible if one looks only at all chambers. COMPUTED

Per cone. Formula Z can cancel: nonzero terms of opposite sign can add to zero. In type A this is completely understood.

Theorem 10 (type A, every cone)**.** *Let τ be an anti-automorphism of P ($n \geq 3$). Call a point odd if it lies on an odd cycle of τ. Then $\chi_{C_P}(-\tau) \neq 0$ if and only if* PROVEN AUDITED

(1) the cycle lengths of τ lie in $\{m, 2m\}$ for one odd squarefree m;

(2) the $\langle \tau \rangle$-orbit of every component K of P has length m or $2m$;

(3) for every component K, the set $K \cup \tau^m K$ contains at most one odd point;

(4) if τ has both odd and even cycles, some odd cycle of τ does not consist of isolated points of P.

[19] notes/vanishing_criterion.md, Verdict table, rows Z and S; audited twice, in notes/audit_vanishing_criterion.md and notes/audit2_vanishing_criterion.md (both: holds). We did not find either statement for proper cones (notes/prior_art_vanishing_criterion.md, §§2–3).

[20] notes/vanishing_criterion.md, row A and §4; audited in both audits named above.

[21] notes/per_cone_vanishing.md, §3; notes/poset_jks_theorem.tex, Lemma 4.1.

[22] notes/prior_art_vanishing_criterion.md, §4.

There is a positive closed formula for $|\chi_{C_P}(-\tau)|$. With Theorem 1 for $+w$, this decides vanishing for every element of $\pm S_n$ on every cone of the braid arrangement.[23]

One ingredient is a small fact about linear extensions: an anti-automorphism t of a poset has a linear extension w with $t(w_i) = w_{n+1-i}$ for all i (a "t-palindromic" extension) if and only if $t^2 = 1$ and t has at most one fixed point.[24] These palindromic extensions are the chambers that $-t$ fixes. The lemma is known in substance: after relabelling, the palindromic extensions are the type-B linear extensions of a B_n poset (Chow; Reiner for signed posets).[25]

PROVEN AUDITED

What is open.

- An exact per-cone vanishing criterion beyond type A: for B_n ($n \geq 5$), D_n, H_3, H_4, F_4 and general arrangements. Off the Coxeter examples, cancellation is common.[26]

  OPEN

- The Crapo-type converse on all chambers. By Crapo's complementation theorem, if the interval $[0, X]$ of the fixed lattice has an element with no complement, then $\mu(0, X) = 0$. On all chambers, the converse held for all 191 flats $X \in Z^*(g)$ tested.[27]

  CONJECTURED

- The Crapo-type converse on cones (Conjecture CR_V): $\chi_V(g) = 0$ if and only if $\Lambda(V)^g$ has an element with no complement. It is false in rank 4. No counterexample was found on the first 18,750 pairs (V, g) tested.[28] A later note gives a counterexample in rank 4: a 12-hyperplane arrangement in $\mathbb{R}^4$ where a cancellation $1 - 1 = 0$ in Formula Z sits on a 9-element lattice in which every element has a complement.[29] So the converse is false in general, but true for posets: on every cone of the braid arrangement (every poset P, every automorphism or anti-automorphism), $\chi_V(g) = 0$ forces an explicit element of $\Lambda(V)^g$ with no complement, in every rank.[30] Moreover, every such zero has an explicit, uniform witness with no complement, given by one of two formulas.[31]

  REFUTED AUDITED

- The converse does hold in rank at most 3 and for pointed cones in every rank: there $\chi_V(g) = 0$ forces an element of $\Lambda(V)^g$ with no complement. In every rank, each zero in which a hyperplane contains $\operatorname{lin} V + \operatorname{Fix} g$ (a "Cor. V" zero) comes with such an element, explicitly.[32]

  PROVEN AUDITED

[23] notes/per_cone_vanishing.md, §3, Theorem TA and row TA±. Audited in notes/audit_per_cone_vanishing.md: "holds with fixes"; the fixes are to proofs and do not change a result, and they have been applied to the source note. A search found no statement of Theorem TA in the literature (notes/prior_art_per_cone_vanishing.md, Verdict table).

[24] notes/per_cone_vanishing.md, §2, Lemma Pal; audited in notes/audit_per_cone_vanishing.md (holds; checked on all 6,847 anti-automorphisms with $n \leq 7$).

[25] notes/prior_art_per_cone_vanishing.md, Verdict table, rows Lemma Pal. The exact "iff" form was not found stated.

[26] Rows PC′ and PC″ of notes/vanishing_criterion.md, and rows B, D, H, F of notes/per_cone_vanishing.md.

[27] notes/vanishing_criterion.md, row CR.

[28] notes/per_cone_vanishing.md, row CR_V and §4. A search found no statement of the conjecture. The converse is false for general finite lattices, and it is known for finite modular lattices (Knop), a class that does not contain the lattices $\Lambda(V)^g$ in general (notes/prior_art_per_cone_vanishing.md, Verdict table).

[29] notes/conjecture_cr_v.md, §0 and Verdict table. Independently audited in notes/audit_conjecture_cr_v.md: the refutation holds; the counterexample was rebuilt from the definitions, and $\chi_V(g) = 0$ was confirmed in four independent ways.

[30] notes/cr_v_type_a.md, Theorem TA-CR; independently audited in notes/audit_cr_v_type_a.md (it holds). The "if" direction is Crapo's corollary. A search found the "only if" direction only for the antichain with an automorphism (Stanley 1982, with Hanlon 1981), not for general posets (notes/prior_art_cr_v_type_a.md).

[31] notes/observation_cc.md, Theorem CC and Cor. U; independently audited in notes/audit_observation_cc.md (it holds).

[32] notes/conjecture_cr_v.md, Theorems 3, P and Q; audited in notes/audit_conjecture_cr_v.md (all three hold).

- In type A, whether $\chi_V(g) = 0$ depends only on $(\operatorname{lin} V, g)$.[33] PROVEN AUDITED
- In general it does not: generic arrangements of four planes in $\mathbb{R}^3$ already give counterexamples.[34] COMPUTED

6 Neighbouring theories

Each paragraph names the analogy and where it stops. Several primary sources could not be opened, and the notes record them as *not accessible*; statements about them below come from secondary accounts.

Orlik–Solomon algebras and Lehrer–Solomon. The Orlik–Solomon algebra is the cohomology of the complement of the complexified arrangement. Lehrer–Solomon describe the S_n-action on it as a sum of modules induced from centralizers. On all chambers, the cone character χ is the top Orlik–Solomon character twisted by det (Orlik–Solomon 1980, Thm. 4.8).[35] The analogy stops at cones. That theory concerns the whole complement, so it sees only all chambers. The cone theory decomposes the space spanned by chambers instead. For all chambers that decomposition matches the cohomology of a different space, Moseley's complement $M_3(\mathcal{A})$ of the codimension-three subspaces, and it differs from the Orlik–Solomon one by a determinant twist on each summand. Nothing in those accounts mentions fixed chambers, cones or lineality spaces.

Lie characters of Coxeter groups. For a reflection group W, the Lie module of the whole arrangement is a W-module, and Aguiar–Mahajan prove a JKS theorem for every arrangement, with the twist carried by an orientation space. This is the all-chambers case of Theorem 5.[36] Bishop, Douglass, Pfeiffer and Röhrle give a necessary support condition for the top Orlik–Solomon component through cuspidal classes, for groups of rank at most 8 and for all chambers only.[37] The cone theory adds proper cones and symmetries outside W, such as $-w$ in type A. The type-A support rule does not generalize in the obvious way: Springer regularity is neither necessary nor sufficient for being nonzero on some cone. The Coxeter element of B_3 vanishes on every cone, and a reflection of B_2 does not.[38] PROVEN AUDITED

Aguiar–Mahajan: Hopf monoids, cones and lunes. The vocabulary here (Tits product, top-lunes, Lie elements as Lune$^{\perp}$) is Aguiar–Mahajan's. In the parts of their books that were read we found no theory attached to a single cone, and one exercise needed correcting: an element is Lie iff its coefficient sum vanishes on every *non-pointed* top-cone, not every non-singleton one.[39] The natural first guess, the left regular band of faces of the cone acting on its own chambers, gives zero for the chain, and on all proper order cones tested. The right object is the ambient Lie PROVEN AUDITED

[33] notes/per_cone_vanishing.md, row TA-L (Cor. 3.5); audited in notes/audit_per_cone_vanishing.md.

[34] notes/per_cone_vanishing.md, row L′ and §4.

[35] notes/prior_art_vanishing_criterion.md, Verdict table, §0.3 and §1; notes/novelty_fixed_point_formula.md, §4. Orlik–Solomon 1980, Thm. 4.8, was read in the source (notes/prior_art_vanishing_criterion.md, §1). Lehrer 1987 and Lehrer–Solomon 1986 were read only through secondary accounts.

[36] notes/novelty_general_theory.md, §5.

[37] notes/prior_art_vanishing_criterion.md, §3.

[38] notes/vanishing_criterion.md, row R; proven by hand given the some-cone criterion, audited in notes/audit_vanishing_criterion.md.

[39] notes/cone_lune_lie_theory.md, §7, audited in notes/audit_cone_lune_lie_theory.md; notes/literature_novelty_jks_maps.md, §4.3.

module restricted to the cone.[40] On the Hopf side, a Hopf monoid of pairs (poset, linear extension), with coproduct by restriction to unions of components, has primitives $\cong \mathsf{Lie}(P)$ in each degree P and a poset-graded Lie bracket, and a local PBW theorem holds.[41] The bracket is not induced by concatenation in the e_ζ model, and a version for general arrangements (via Aguiar–Mahajan's bimonoids) has not been worked out.

COMPUTED

PROVEN AUDITED

Whitehouse modules and Lie in type B. The Whitehouse module is a module for the larger group S_{n+1} whose restriction to S_n is Lie_n (classical background, not discussed in the notes); type-B analogues of the free Lie algebra (Bergeron; Gottlieb–Wachs; Aguiar–Mahajan §14.9) are modules for the hyperoctahedral group on the B_n arrangement. The analogy is "extra symmetry": the cone theory sees the group generated by $\mathrm{Aut}(P)$ and the anti-automorphisms acting as $-\tau$, which is a subgroup of $\pm S_n$, and the $-w$ rule above is a statement about it. It stops at the arrangement: type-B Lie characters are traces on the B_n arrangement, while $-w$ here acts on cones of the A_{n-1} arrangement. Whether some type-B Lie character has the same support as the $\{m, 2m\}$ rule was not checked, and a search for Whitehouse modules together with linear extensions found nothing relevant.[42] Whether H_P carries a hidden action of a group larger than $\mathrm{Aut}(P)$, in the spirit of the Whitehouse module, is not studied in the notes.

Hanlon's fixed-point Möbius computations. Hanlon computed $\mu(\Pi_n^g)$ using the *hinge*, the partition of $[n]$ into cycles of g, and its complements. The proof of Theorem 1 applies Weisner's theorem at the same element (the orbit partition O_g), and for the antichain its set $F(P, g)$ is exactly Hanlon's complements. On all chambers of the braid arrangement, Formula Z is Hanlon's count of complements in substance.[43] The analogy stops at the lattice. Hanlon's method uses supersolvability and geometric lattices, and $\mathcal{O}(P)^g$ is neither in general. Weisner plus Möbius inversion needs neither. We did not find the poset formula of Theorem 1 in the literature. Whether it follows from Baclawski–Björner's work on fixed points and complements is open, because that paper could not be read.

Oriented matroids. As explained above, the wedge theorem, the Zaslavsky count and the cone JKS theorem hold for T-convex sets of topes of any simple oriented matroid. They do not need realizability, because Brown's theory of left regular bands is stated for any finite left regular band.[44] The notes state the fixed-point formula and the vanishing results only for real arrangements with linear symmetries, and the sections on more general objects (complexes of oriented matroids) have not been audited.

7 Why it might matter

7.1 The motivation: moonshine

Monstrous moonshine begins with a numerical coincidence. The coefficient 196884 of the modular j-function is $1 + 196883$, the sum of the two smallest dimensions of irreducible representations of

[40] notes/cone_lune_lie_theory.md, Verdict row 3; proven for the chain, computed for $n \leq 5$.

[41] notes/general_theory.md, §2, Theorems 6–8; audited in notes/audit_general_theory_hopf.md (hold, with wording fixes).

[42] notes/prior_art_vanishing_criterion.md, §4 and §0.3 (Bergeron 1991 not accessible); notes/literature_novelty_jks_maps.md, §1.4 (query log).

[43] notes/novelty_search_2.md, §§2.1–2.2; notes/prior_art_vanishing_criterion.md, §2.1.

[44] notes/audit_oriented_matroid_cones.md, §1.

the Monster group. McKay, Thompson, Conway and Norton conjectured that each coefficient of j is the dimension of a graded piece of a Monster module, and that replacing dimensions by traces of each group element gives other modular functions. Borcherds proved this. An important step was a Lie algebra with Monster symmetry whose denominator identity is a product formula of Witt type.

This program was started by asking whether something analogous could exist for the alternating groups. The research note that started it describes a chain of the same shape:[45]

$$\begin{aligned}&\text{a large combinatorial family} \;\to\; \text{equivariant homology}\\ &\quad\to\; \text{Lie characters} \;\to\; \text{Witt / plethystic-logarithm identities.}\end{aligned}$$

For posets, the last two arrows are now theorems. Sum $\mathsf{Lie}(P)$ over all labelled posets on $[n]$ to get an S_n-module V_n. Let a_n count naturally labelled posets, $A(x) = 1 + \sum_n a_n x^n$, and let $b_n = n[x^n] \log A(x)$.

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Theorem 11. *b_n counts the labelled posets having a rotation of $12\cdots n$ as a linear extension; $V_n \cong \mathrm{Ind}_{C_n}^{S_n}(\zeta \otimes \mathbb{C}[B_n])$ for that set B_n; and*

$$\operatorname{ch} V_n = \frac{1}{n}\sum_{d\mid n} \mu(d)\, b_{n/d}\, p_d^{n/d}, \qquad \sum_{n\geq 1} \operatorname{ch} V_n = \sum_{d\geq 1} \frac{\mu(d)}{d} \log A(p_d).$$

The global homology module is $\operatorname{sgn} \otimes V_n$.[46]

Compare Witt's formula $\operatorname{ch} \mathrm{Lie}_n = \frac{1}{n}\sum_{d\mid n} \mu(d) p_d^{n/d}$: the single generator is replaced by the series of naturally labelled posets. Restricted to even permutations the sign twist disappears, so on $\mathrm{Aut}(P) \cap A_n$ the homology H_P is simply $\mathsf{Lie}(P)$.

No poset-specific modular forms have been found. This is a motivation, not a result. The point of the general theory, from this angle, is a precondition: before looking for an extra grading whose traces could be modular, one needs a Lie theory for posets that is coherent, with modules, characters, functoriality and global generating functions that fit together. The results above supply the modules, the characters and the global generating function. Functoriality is only partial: an isomorphism natural along order extensions exists exactly on connected posets, and no canonical bracket on the e_ζ model is known (the commutator of concatenations does not descend to it).[47]

A first scoping search for such a grading has been made.[48] It tested four candidates: the global module V_n graded by n, g-fixed P-partitions of a finite poset, P-partitions of some infinite posets, and a Fock space built on $\mathsf{Lie}(P)$. The only modularity it found comes from Frame shapes (eta products), which arise from the character of any rational representation of any finite group, so they say nothing specific about posets. The poset-specific series either grow too fast to converge anywhere (V_n) or are rational functions of q, and the note proves that neither kind can be modular. Restricting to the alternating group changes nothing. Its recommendation is to stop this line.

An independent audit re-derived those proofs and found no error that reverses a verdict, but it confirmed the stop only on a *narrower space* than the note claimed.[49] That space is exact modular

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[45] notes/poset_jks_research_note.tex, section "Relation to the moonshine motivation".

[46] notes/poset_jks_theorem.tex, §7: Prop. 7.1, Cor. 7.2, Theorem 7.3 and Prop. 7.4(a)–(c); audited in notes/audit_cyclic_compatible_posets_bn.md and notes/audit_character_identity_proof.md. The sign-twisted generating function of Prop. 7.4(d) has been computed but has had no separate audit.

[47] notes/poset_jks_theorem.tex, §6, Theorem 6.8, and §9, items 1–2.

[48] notes/moonshine_scoping.md, Verdict table and §4.

[49] notes/audit_moonshine_scoping.md, Verdict and §1: "stop holds on a narrower space".

transformation laws for graded traces built from the free poset Hopf monoid and its sub-Hopf monoids, from finite objects attached to a single poset together with the Fock spaces of $\mathsf{Lie}(P)$, and from P-partitions of three specific infinite posets. Three things were left open: mock modularity of the rational series, unlabelled series of certain component-closed classes of posets, and other infinite posets. The note's broader claim, that every natural grading falls into one of its cases, was not established.

A follow-up note takes up those three open items and claims that the stop still holds: every natural class it tests is excluded, mock modularity is excluded for cuspidal shadows, and the one escape left is a multi-cusp case for $\mathbb{N}^k$ ($k \geq 3$) that no moonshine formulation needs. That note has *not* been independently audited.[50] So the honest summary is: the stop is audited on a stated space of candidates; outside that space it rests on unaudited arguments; and nothing found so far points to a poset-specific modular object.

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7.2 A second measure: counting and structure of linear extensions

Does the theory help to count linear extensions, or to understand their structure? Here is where it touches them, in what the notes prove.

- $e_C(P)$ counts the rotation classes of linear extensions, and $\mu(\mathcal{O}(P)) = (-1)^{n-1} e_C(P)$. A count becomes a Möbius function, and the proof needs no topology.
- The cone Zaslavsky theorem writes $|\operatorname{Lin}(P)|$ as a sum of products of e_C over partitions of the components of P.
- The character of $g \in \operatorname{Aut}(P)$ is a sum over the rotation classes of linear extensions fixed by g, weighted by roots of unity. The closed formula says this sum is $\mu(o)$ times a count of e_C's of blocks.
- For anti-automorphisms, the fixed chambers are the palindromic linear extensions, and Lemma Pal says exactly when they exist.

A separate note asks these questions directly, and compares rotation with *promotion*, the classical cyclic action on linear extensions (Schützenberger; Stanley).[51] Write T for the components of P and $e(T) = |\operatorname{Lin}(T)|$.

PROVEN
AUDITED

- *Counting.* $e_C(P) = \prod_T e(T) \cdot e_C(\bigsqcup_T C_{|T|})$, where C_k is a k-element chain. So e_C carries no information beyond the $e(T)$ and the sizes of the components, and for connected P it is $e(P)$. The notes find no new recursion or bound for $e(P)$.
- *Promotion on disjoint unions.* Record a linear extension by the word saying which component each letter comes from, together with its restrictions to the components. Promotion rotates that word by one place and promotes inside the component of the first letter.
- *Orbits and classes.* The number of promotion orbits is at most $e_C(P)$, for every P. If promotion of each component T has order dividing $|T|$, then each rotation class lies inside one promotion orbit; without that hypothesis this can fail. Promotion orbits equal rotation classes exactly when P is a chain or an antichain.

[50] notes/moonshine_residual.md, Verdict table.

[51] notes/linear_extensions.md, §§1–5, audited in notes/audit_linear_extensions.md: "holds with fixes". The fixes change wording, scope and citations only, and no result. The statements here are the audit's corrected ones.

- *The Lie module does not see promotion.* Promotion descends to $\mathsf{Lie}(P)$ exactly when P is connected or an antichain.
- *Cyclic sieving.* Promotion cyclic sieving is closed under disjoint union. The resulting polynomial is the q-multinomial coefficient of the component sizes times the product of the components' polynomials. The proof uses a theorem of Reiner, Stanton and White (cited).[52]
- *A canonical grading.* Let R_P be the span of the lunes inside C_P, so that $\mathbb{C}[\operatorname{Lin} P]/R_P \cong H_P \otimes \operatorname{sgn}$ by Theorem 5. Then $\mathbb{C}[\operatorname{Lin} P]/R_P$ splits as $\mathbb{C}[\prod_T \operatorname{Lin} T] \otimes Q$, where Q is a shuffle quotient that depends only on the sizes of the components; $\operatorname{Aut}(P)$ acts diagonally.

None of these uses the homology H_P or the Lie module in an essential way: they use only the combinatorics of cuts. Neither H_P nor $\mathsf{Lie}(P)$ carries a grading or a cyclic action that yields a q-analogue; we found none (promotion does act on $H_P \otimes \operatorname{sgn}$ in the two-chain case of Theorem 12 below, but that gives no new q-analogue). By computer, the relations R_P are homogeneous for descents or major index on none of the 108 disconnected posets with $n \le 6$ (one natural labelling each). Among the posets with $n \le 7$ whose promotion has order dividing n, the major-index polynomial fails to be a cyclic sieving polynomial for 49.[53]

COMPUTED

Question 1. The notes find no new way to count $|\operatorname{Lin}(P)|$, which is hard in general. Is there an application of the Lie theory to counting that does not reduce to the components?

When does promotion act on the homology model? The *Lie module* bullet above says when promotion descends to $\mathsf{Lie}(P)$. The same question for the other model, $\mathbb{C}[\operatorname{Lin} P]/R_P$, is now settled.

PROVEN
AUDITED

Theorem 12 (promotion on R_P)**.** *Let P be a disconnected finite poset. Then promotion maps R_P into itself, and so acts on $\mathbb{C}[\operatorname{Lin} P]/R_P$, if and only if P has exactly two connected components and both are chains.*[54]

For the two-chain poset $C_a \sqcup C_b$ the reason is simple: R_P is the line spanned by the sum of all linear extensions, and promotion only permutes them. For every other disconnected poset the note writes down one explicit generator r of R_P and one explicit Lie element that pairs to 1 with (a graded piece of) the promoted vector but to 0 with all of R_P; that certifies that promotion leaves R_P.

Put together with the Lie-module bullet: for connected P promotion acts on both models. For disconnected P it acts on $\mathsf{Lie}(P)$ exactly when P is an antichain, and on $\mathbb{C}[\operatorname{Lin} P]/R_P$ exactly when P is two chains. For $n \ge 3$ the second model is $H_P \otimes \operatorname{sgn}$; the only poset in both families is $C_1 \sqcup C_1$, where $n = 2$ and that identification is not available.[55]

[52] Prior art, from notes/prior_art_linear_extensions.md, Verdict table and "Bottom line". The description of promotion on disjoint unions is known for two chains (Stanley; Striker–Williams) and for any number of chains and for rectangles (Alexandersson–Kantarci Oğuz–Linusson); its mechanism is Stanley's, and the general statement was not found. The closure of cyclic sieving is known for chains (Reiner–Stanton–White, Prop. 4.4) and for rectangles (Alexandersson–Kantarci Oğuz–Linusson, Thm. 37), and Hopkins's survey treats it as evident; the general form was not found stated or proved. The first two statements of the orbits bullet were not found.

[53] notes/linear_extensions.md, Verdict table and §3, as corrected by notes/audit_linear_extensions.md, F6–F7.

[54] notes/promotion_on_R_P.md, §2, Theorem 1. The "if" direction is in notes/linear_extensions.md, §2. Audited in notes/audit_promotion_on_R_P.md: "holds with fixes"; the fixes are about precision and wording, change no result, and have been applied to the source note. Also checked by computer on every disconnected poset with $n \le 8$: 16 of 2990 have the property, exactly the two-chain ones. Not found in a brief search (four web searches, recorded in the audit, §6).

[55] notes/promotion_on_R_P.md, Verdict table, with notes/linear_extensions.md, Prop. D.

7.3 What the theory says about counting linear extensions

The counting question of the previous subsection has one positive answer, at the level of *refined* counts. Recall Stanley's *order polynomial*: $\Omega(P,m)$ is the number of order-preserving maps $f : P \to \{1,\dots,m\}$ (so $i <_P j$ implies $f(i) \le f(j)$). Fix a natural labelling ($i <_P j$ implies $i < j$). Then

$$\Omega(P,m) = \sum_{w \in \mathrm{Lin}(P)} \binom{m+n-1-\mathrm{des}(w)}{n},$$

where $\mathrm{des}(w)$ is the number of descents of the word w. So the order polynomial knows $|\mathrm{Lin}(P)|$ (its leading coefficient is $|\mathrm{Lin}(P)|/n!$) and the descent distribution on $\mathrm{Lin}(P)$. The notes check that the face numbers of the cone C_P are exactly this classical information.[56]

CLASSICAL (CITED)

Reading $\mathcal{O}(P)$ from the top instead of the bottom gives something new. Define the *cyclic order polynomial*

$$\Omega_C(P,m) = \sum_{\pi \in \mathcal{O}(P)} |\mu(\pi,\hat{1})| \binom{m}{|\pi|},$$

where $|\pi|$ is the number of blocks.

Theorem 13 (Theorem Ω)**.** *Fix a connected component T of P. Then $\Omega_C(P,m)$ is the number of order-preserving maps $f : P \to \{1,\dots,m\}$ such that every nonempty set $\{x : f(x) \ge v\}$ that is a union of components of P contains T. Its leading coefficient is $e_C(P)/n!$, and for connected P it is the order polynomial $\Omega(P,m)$.*[57]

PROVEN AUDITED

Its h-vector, the analogue of the descent distribution, is a count of rotation classes. Call a natural labelling *separated* if every minimal element of a connected component with at least two points has a smaller label than every maximal element of such a component; separated labellings always exist. The *cyclic descents* of a word are its descents together with position n if $w_n > w_1$; their number does not change when the word is rotated.

Theorem 14 (Theorem H)**.** *Let $n \ge 2$ and choose a separated natural labelling. Then*

$$\Omega_C(P,m) = \sum_{\kappa} \binom{m+n-\mathrm{cdes}(\kappa)}{n},$$

the sum running over the $e_C(P)$ rotation classes κ of words that contain a linear extension, where $\mathrm{cdes}(\kappa)$ is the number of cyclic descents. In particular the h-vector of Ω_C is the distribution of $\mathrm{cdes}-1$ on those classes, so it is nonnegative and sums to $e_C(P)$.[58]

PROVEN AUDITED

Example 2. Take P to be $1 < 2$ together with an isolated point 3, the second poset of Figure 2; its labelling is separated. All five set partitions are order-preserving, with $|\mu(\pi,\hat{1})| = 1$ for $\hat{1}$ and for the three two-block partitions, and $e_C(P) = 2$ for the bottom. So $\Omega_C(P,m) = m + 3\binom{m}{2} + 2\binom{m}{3}$, which is 5 at $m = 2$. Directly: of the six order-preserving maps to $\{1,2\}$, only $f(1) = f(2) = 1$, $f(3) = 2$ violates the condition with $T = \{1,2\}$, since its top level set $\{3\}$ is a component that misses T. The rotation classes are $\{123, 312\}$, with one cyclic descent, and $\{132\}$, with two. Theorem H gives $\binom{m+2}{3} + \binom{m+1}{3}$, which is again 5 at $m = 2$.

[56] notes/linext_statistics.md, §1, Prop. 1; Stanley, *Enumerative Combinatorics* vol. 1, §3.15.

[57] notes/linext_statistics.md, §2.1; audited in notes/audit_linext_statistics.md (it holds; the audit re-derived the proof and checked the identity as polynomials on all 405 posets with $n \le 6$). The count does not depend on T.

[58] notes/h_vector_omega_c.md, §2; audited in notes/audit_h_vector_omega_c.md (it holds, with two wording fixes, applied). Checked by computer, from the Möbius definition, on all 2450 posets with $n \le 7$. Without separation the statement can fail.

The separation condition is exactly what is needed, in the following sense. Take any natural labelling. Read each rotation class as a cyclic word, and call a place between two cyclically adjacent letters a *junction* if cutting the word there gives a linear extension. Count each class by the least number of descents among its linear extensions. Then this count has the h-vector of Ω_C as its distribution if and only if no rotation class with at least two linear extensions has all its junctions ascending (the smaller letter before the larger).[59]

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For the antichain, Theorem H says that the cyclic descents of cyclic permutations follow the Eulerian distribution of S_{n-1}; this case is known (Petersen, after Fulman).[60] For general P, the searches recorded in the notes did not find Theorem Ω, its descent expansion, or Theorem H. A stronger statement also holds: a quasisymmetric lift of Ω_C expands positively in Gessel's fundamental basis. It is essentially the fundamental lemma of P-partitions of Stanley and Gessel, with the anchor component labelled last.[61] Stanley's reciprocity carries over: $(-1)^n\Omega_C(P,-m)$ counts the *strictly* order-preserving maps to $\{1,\dots,m\}$ in which no union of components missing T takes only values at least as large as every value outside it, and turning P upside down changes nothing, $\Omega_C(P^*,m)=\Omega_C(P,m)$.[62] The polynomial identity is a form of Stanley's reciprocity, which holds one linear extension at a time; only its reading as this anchored count of strict maps was not found in the literature. So the Lie theory does not give a new way to compute $|\operatorname{Lin}(P)|$, but its Möbius function does give a new refined count of the rotation classes it is built on. One can also ask exactly when the h-vector of Ω_C reads the same backwards. Call P *graded* if all its maximal chains have the same number of elements. For $n\geq 2$, the h-vector is palindromic if and only if either P is connected and graded, or no component of P contains a chain of three elements.[63] For connected P this is classical, since there Ω_C is Stanley's order polynomial (Stanley, *Enumerative Combinatorics* vol. 1, Cor. 3.15.18(b)). The case of disconnected P is proved in these notes. Being graded component by component is not enough: two disjoint 3-element chains give the h-vector $(1,8,9,1)$.

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7.4 Cyclic sieving on rotation classes

Rotation and symmetry also meet in a second way. Write $\operatorname{Cyc}(P)$ for the set of rotation classes of words that contain a linear extension, so $|\operatorname{Cyc}(P)|=e_C(P)$. Group the components of P into isomorphism types, type T_i occurring k_i times. Permuting the copies of each type gives a group $W=\prod_i S_{k_i}$ of automorphisms of P, and it permutes $\operatorname{Cyc}(P)$. A polynomial $X(q)$ *cyclically sieves* an element g if, for every j, evaluating X at the jth power of a primitive $o(g)$th root of unity gives the number of classes fixed by g^j. At $q=1$ this is the count $X(1)=e_C(P)$.

The natural candidate is the *fake-degree polynomial* $X_P(q)$ of $\operatorname{Cyc}(P)$ as a W-set. It has

[59] notes/h_double_prime.md, Theorem H″ with Theorem H′ of notes/h_vector_omega_c.md; audited in notes/audit_h_double_prime.md (it holds). A separated labelling never has such a class. Example: $1<2$, $4<5$ and an isolated point 3. The class of 41523 contains the linear extensions 41523 and 34152, its junctions $2\to 3$ and $3\to 4$ both ascend, and the count gives $(1,9,8,2)$, while the h-vector is $(1,9,9,1)$. Checked by computer, from the Möbius definition, on all 101658 naturally labelled posets with $n\leq 7$.

[60] Prior art, from notes/prior_art_linext_statistics.md, Verdict table: the antichain case is T. K. Petersen, *Cyclic descents and P-partitions*, Prop. 1.1, credited there to Fulman. The statistic is Cellini's cyclic descent number. Ω_C is not a toric order polynomial in the sense of Develin–Macauley–Reiner or Adin–Gessel–Reiner–Roichman, and its h-vector is not a cyclic-descent-extension distribution in the sense of Adin–Reiner–Roichman.

[61] notes/qsym_refinement_omega_c.md, Theorem Q and Corollary F+; audited in notes/audit_qsym_refinement_omega_c.md.

[62] notes/reciprocity_omega_c.md, Theorem R and Prop. ⋆; audited in notes/audit_reciprocity_omega_c.md; prior art in notes/prior_art_reciprocity_omega_c.md (Stanley, *Enumerative Combinatorics* vol. 1, Lemma 3.15.9).

[63] notes/palindromic_h_c.md, Theorem S′; audited in notes/audit_palindromic_h_c.md. Checked by computer on all 19,448 posets with $2\leq n\leq 8$.

nonnegative integer coefficients and $X_P(1) = e_C(P)$.[64]

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Theorem 15 (a known phenomenon that shows up here)**.** *Let $P = T^{\sqcup k}$ be k copies of a connected poset T, and let σ rotate the copies. Then X_P cyclically sieves σ. More generally X_P cyclically sieves every Springer-regular element of W.*[65]

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What is particular to posets is what comes next. None of it was found in the literature searched, which is a search result and not a proof of novelty.[66]

- *A closed form.* For $P = T^{\sqcup k}$, with $c_j = e_C(T^{\sqcup j})$ and φ Euler's function,

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  $$X_P(q) = \sum_{d|k} \frac{\varphi(d)}{d} \cdot \frac{c_{k/d}}{(k/d)!} \cdot \frac{(q;q)_k}{(1-q^d)^{k/d}}.$$

  For $k = 2$ it reads $X_P = \big((c_2 + c_1) + (c_2 - c_1)q\big)/2$.[67]

- *Exactly which elements of W (Theorem C).* For $g \in W$ of order N, consider: (a) every divisor $e > 1$ of N is at most $\max k_i$; (b) if such an e divides every k_i, then $g^{N/e}$ moves the copies in cycles all of length e. Then X_P sieves g *if and only if* (a) and (b) hold. The hard part is that (a) is needed even when $\gcd(k_i) > 1$. It comes from a non-vanishing theorem: if $e \geq 6$ and $e > \max k_i$, then $X_P(e^{2\pi i/e}) \neq 0$, with one exception, the three-point antichain at $e = 6$, where $X_P = 1 + q^3$. That exception never matters, since no symmetry of three copies has order 6.[68]

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- *No single polynomial for all symmetries.* For the four-point antichain A_4, a transposition fixes no class and a double transposition fixes two, though both have order 2. A polynomial has one value at $q = -1$, so no single X serves every cyclic subgroup of $\mathrm{Aut}(A_4)$. No poset with $n \leq 3$ is such an obstruction, and A_4 is the only one with $n = 4$.[69]

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Other natural candidates (major index or inversions on representatives, a q-version of the component formula for e_C) all fail to sieve.[70]

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[64] notes/cyclic_sieving_cyc.md, §3, Theorem A; audited in notes/audit_cyclic_sieving_cyc.md ("holds with fixes"; the fixes concern counting of the evidence, citations and wording, and change no theorem). Theorem A is standard (notes/prior_art_cyclic_sieving_cyc.md, Verdict table).

[65] notes/cyclic_sieving_cyc.md, Verdict table and Corollary T; audited in notes/audit_cyclic_sieving_cyc.md. This is an instance of a published theorem, not a new one: Westbury (2016), Cor. 2.6, for the rotation, and Reiner–Stanton–White, Thm. 8.2 (or Sagan's survey, Thm. 8.2) for regular elements, applied orbit by orbit (notes/prior_art_cyclic_sieving_cyc.md, Verdict table and §1; the theorem number was checked in the published paper).

[66] notes/prior_art_cyclic_sieving_cyc.md, Verdict table and "Bottom line". Springer's 1974 paper and Lehrer–Michel (2003) were then read (notes/prior_art_cyclic_sieving_cyc_2.md). The proof of the vanishing behind Theorem C is Springer's argument, used at roots of unity where his theorem does not apply; the general form of that argument (Prop. E there) was audited in notes/audit_prior_art_cyclic_sieving_cyc_2.md. No statement of the closed form, of Theorem C beyond regular elements, or of the A_4 obstruction was found. A further search (notes/prior_art_cyclic_sieving_gcd.md) did not find the necessity of clause (a) when $\gcd(k_i) > 1$, or the non-vanishing theorem behind it. The Lehrer–Taylor book chapter on eigenspace theory could not be obtained and was not read.

[67] notes/cyclic_sieving_cyc.md, §3, formula ($\star$); audited (re-derived, and all 23 posets $T^{\sqcup k}$ with $n \leq 8$ reproduced).

[68] notes/cyclic_sieving_cyc.md, §3, Theorem C and Prop. 3; audited in notes/audit_cyclic_sieving_cyc.md, §§2–3 (which also supplied the $\gcd(k_i) = 1$ case and corrected the count of tested classes). The case $\gcd(k_i) > 1$ is notes/cyclic_sieving_gcd.md, Theorem G and Corollary H; its independent audit (notes/audit_cyclic_sieving_gcd.md) finds every claim holds, with no fixes, and re-checks the one computer-assisted step (490 cases, exact arithmetic) by a different method. The proof uses that every component has at least one linear extension: for a formal value of that count below 1 the conclusion fails. For elements all of whose powers are regular, Theorem C is the known

7.5 What the general theory buys over the poset theorem

- *Unification.* Classical JKS, Folkman, Zaslavsky, Jenča–Sarkoci, Hanlon's values of $\mu(\Pi_n^g)$ and the component formula become special cases of three statements: the wedge theorem, the cone JKS theorem, and the fixed-point formula.

- *Computations.* The fixed-point formula was checked on 7,200 pairs (cone, symmetry) in types A_2–A_5, B_2–B_4, D_4 and H_3.[71] The equivariant cone JKS theorem was checked on proper cones, and the determinant twist is visible there: without it the identity fails on 9 of the 33 cones of B_2.[72] COMPUTED

- *The vanishing question.* The poset theorem alone does not suggest it. The cone theory answers "nonzero on some cone" exactly, answers "nonzero on this cone" in type A, and offers a precise Crapo-type conjecture for the rest. On cones that conjecture fails in rank 4, and holds in rank at most 3 and on every cone of a poset (all audited).

What is still open. The exact per-cone vanishing criterion outside type A; the Crapo-type converse on all chambers (conjectured) and the right per-cone criterion in rank ≥ 4, now that the converse on cones is refuted there; a canonical bracket or cooperad on the e_ζ model; a version of the Hopf-monoid picture for general arrangements; and a Lean formalization of anything beyond the poset theorem.

8 Status of the Lean formalization

What "formalized" means. Lean is a proof assistant: a program that accepts a definition or a proof only if every step follows, by its own small set of logical rules, from earlier steps. "The poset theorem is formalized" means that its statement has been written out in Lean, using the Mathlib library, and a complete proof has been checked by the computer, down to the axioms. A proof can cheat in two ways: by leaving a hole marked `sorry`, or by assuming an extra axiom. So the check is that the build succeeds, that no `sorry` remains, and that Lean's own `#print axioms` lists only the three standard axioms of Lean's logic. What the computer cannot check is whether the Lean statements say the same thing as the mathematics; that was reviewed separately, by comparing the definitions and the main statements with the paper.

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phenomenon above; the part beyond regular elements is what was not found (its proof uses Springer's evaluation argument at non-regular roots of unity; notes/prior_art_cyclic_sieving_cyc_2.md).

[69] notes/cyclic_sieving_cyc.md, Verdict table, Prop. O′; audited (by hand, and every poset with $n \leq 6$).

[70] notes/cyclic_sieving_cyc.md, Verdict table, last row; reproduced in the audit, with a remark that two of the counts depend on the labelling.

[71] notes/fixed_point_formula.md, Verdict table.

[72] notes/general_theory.md, Summary item 4.

Lean status, as of 31 October 2026 (from lean/PLAN.md and notes/lean_verification.md). Lean 4 with Mathlib.

- **Complete: no `sorry` remains.** Every stated theorem is proved: the character identity of Theorem 1 and its closed formula, and the isomorphism $H_P \cong \mathrm{sgn} \otimes \mathsf{Lie}(P)$ (`main_iso`). The three inputs that the paper cites are proved too: the Jenča–Sarkoci theorem in its homological form, P. Hall's theorem, and the Hopf trace formula. So are Theorems S(a), S(b), Ψ and Cor. Ψ.1, the explicit isomorphism $\mathsf{Lie}(P) \cong \mathbb{C}[\mathrm{Lin}\, P]/R_P$.
- **Standard axioms only.** Every result listed by `#print axioms` depends only on `propext`, `Classical.choice` and `Quot.sound`.
- **Independently confirmed.** A fresh clone of the repository was built from scratch: the build succeeded, with no `sorry` and only the standard axioms, and the statements are the ones the faithfulness review checked, with one intended change (a hypothesis $n > 0$ added to three statements that were false for $n = 0$, as the review proposed). Positive controls (a deliberately planted `sorry`) were detected. Verdict: CONFIRMED.
- The cone, oriented-matroid, fixed-point, vanishing, promotion and cyclic sieving results are *not* formalized.

Where to read more

notes/poset_jks_theorem.tex	the poset theorem, its proofs, the global module
notes/general_theory.md	cones: wedge, Zaslavsky, cone JKS, Hopf monoids
notes/oriented_matroid_cones.md	the oriented-matroid version
notes/fixed_point_formula.md	characters on cones
notes/vanishing_criterion.md	Formula Z, some-cone criterion, type A
notes/per_cone_vanishing.md	per-cone vanishing, Crapo, Conjecture CR_V
notes/conjecture_cr_v.md	refutation of CR_V in rank 4 (audited)
notes/linear_extensions.md	promotion, cyclic sieving, counting
notes/promotion_on_R_P.md	when promotion acts on $\mathbb{C}[\mathrm{Lin}\, P]/R_P$
notes/linext_statistics.md	statistics of $\mathrm{Lin}\, P$; Theorem Ω (audited)
notes/h_vector_omega_c.md	Theorem H: the h-vector of Ω_C (audited)
notes/cyclic_sieving_cyc.md	cyclic sieving on rotation classes
notes/moonshine_scoping.md	the moonshine scoping (audited, narrower space)
notes/moonshine_residual.md	the residual moonshine items (not yet audited)
notes/prior_art_vanishing_criterion.md	what was searched and found
notes/novelty_*.md	the same, for the other results
notes/citation_check.md	cited statements checked against sources
lean/PLAN.md	the formalization
notes/lean_verification.md	its clean-checkout verification